



Guidelines

Digoxin

Scope

Site	Service/Department/Unit	Disciplines
SCGH and OPH (i.e. SCGOPHCG)	Organisation Wide	Medical, Nursing, Pharmacy

Introduction

This guideline provides guidance to all Sir Charles Gairdner and Osborne Park Health Care Group (SCGOPHCG) staff involved in the prescription, administration, supply and management of patients on digoxin. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General Information

Digoxin slows the heart rate and AV nodal conduction by increasing vagal impulses and reducing sympathetic activity. Digoxin increases the force of contraction of the heart by inhibiting sodium-potassium ATPase. Inhibiting sodium potassium exchange increases intracellular calcium availability and release resulting in an increased force of myocardial contraction.

2.0 Presentation

- Digoxin 62.5microg tablet (Sigmaxin®, Lanoxin®)
- Digoxin 250microg tablet (Sigmaxin®, Lanoxin®)
- Digoxin oral liquid 50microg/mL (only for patient who are unable to tolerate tablets/capsules)
- Digoxin 500microg/2mL ampoule for injection (Lanoxin® adult)

3.0 Indication

- Control of ventricular rate in atrial fibrillation and atrial flutter.
- Treatment of congestive heart failure
- Refer to the [WA Statewide Medicines Formulary \(SMF\)](#) for full details of restriction

4.0 Contraindications¹

- Second degree or complete heart block (without pacemaker)
- Ventricular tachycardia and ventricular fibrillation
- Supraventricular arrhythmias involving accessory pathway (Wolf-Parkinson-White syndrome)
- Hypertrophic obstructive cardiomyopathy
- Cor pulmonale
- Constrictive pericarditis
- Hypersensitivity to digoxin, other digitalis glycosides, or any other constituent

5.0 Precautions¹

- Carotid sinus hypersensitivity:
 - digoxin may cause an increase in vagal tone
- Electrolyte imbalances:
 - Hypokalaemia & hypomagnesaemia may increase toxicity risk.
 - Hypercalcaemia & hyperkalaemia increase risk of digitalis induced arrhythmias. With hypocalcaemia the use of digoxin may be ineffective
- Thyroid dysfunction:
 - Hyperthyroid patients may be less sensitive requiring larger doses, while hypothyroid patients may be more sensitive and require smaller doses
- Idiopathic hypertrophic subaortic stenosis:
 - may worsen left ventricular outflow restrictions
- Ischaemic heart disease, acute myocardial infarction, acute myocarditis, myxoedema, severe pulmonary disease and DC cardioversion:
 - increased myocardium sensitivity to digoxin and increased risk digoxin induced arrhythmias
- Impaired renal function:
 - Increased risk of toxicity due to reduced excretion of the drug
- Sick sinus syndrome:
 - risk of sinus bradycardia or sinoatrial (SA) block
- Cardiac amyloidosis:
 - avoid use in amyloidosis associated heart failure. Extreme caution should be exercised in situation where digoxin is needed to control the ventricular rate in patients with atrial fibrillation and cardiac amyloidosis
- Myocarditis:
 - digoxin rarely causes vasoconstriction - avoid use in patients with myocarditis

6.0 Drug Interactions²

- There are a number of important drug interactions with digoxin, consult [SCGOPHCG approved references](#) (e.g. Australian Medicines Handbook) for more information.
- Digoxin can cause arrhythmias; administration with other drugs that can also cause arrhythmias may increase this risk.
- Digoxin slows cardiac conduction and causes bradycardia; administration with other drugs that also have this effect increases the risk of significant bradyarrhythmia; monitor heart rate and adjust doses if needed.
- Electrolyte abnormalities, e.g. hypercalcaemia, hypokalaemia, increase the risk of arrhythmias, and may also prolong the QT interval.
- Giving digoxin (a P-gp substrate) with inhibitors or inducers of this protein may increase the risk of its adverse effect or decreased its efficacy. Monitor digoxin concentration and adverse effects when there is any change to drug treatment.

7.0 Dosage

Note: Digoxin may have a proarrhythmic effect¹

7.1 Loading dose

A loading dose may be used for more rapid control of ventricular rate in AF and atrial flutter. Digoxin loading can usually be achieved by the oral route.

- Adults:

- Oral/IV: 250 to 500 micrograms, every 4 to 6 hours according to response up to a total of 1.5 mg (1500 micrograms)
- Older adults:
 - Oral/IV: 125 to 250 micrograms every 4 to 6 hours according to response, up to a total of 500 micrograms
- Renal impairment:
 - Creatinine clearance (CrCL): <60 mL/minute, halve dose

7.2 Maintenance dose

- Adults:
 - Oral: 125 to 250 micrograms once daily (rarely increased up to 500 micrograms daily)
- Older adults:
 - Oral: 62.5 to 125 micrograms once daily
- Renal impairment:^{1,2} Reduce dose according to drug level if necessary
 - CrCl 30–60 mL/minute: oral 62.5 to 250 micrograms once daily
 - CrCl 10–30 mL/minute: oral 62.5 to 125 micrograms once daily
 - CrCl <10 mL/minute: oral 62.5 micrograms once daily or on alternate days

8.0 Intravenous (IV) administration

If giving intravenously, IV Infusion is the preferred method of administration

IV administration effects occur within 5 to 30 minutes compared with oral administration effects occurring within 30 to 120 minutes.²

- IV Infusion
 - Dilute the dose in 50–100 mL of 0.9% sodium chloride or 5% glucose and infuse over 2 hours. **Loading doses can be infused over 10 minutes.** Use an infusion pump.³
- IV Injection
 - May be injected undiluted over 10 minutes in fluid restricted patients

NOTE: Rapid intravenous injection can cause vasoconstriction producing hypertension and/or reduced coronary flow ^{1,3}

9.0 Monitoring^{1,2,3}

9.1 Administration monitoring requirements:

- In non-cardiac monitored patients: Monitor and record heart rate and blood pressure immediately prior to start of infusion and at 10 minutes intervals until infusion ceases; and 1 hour post infusion
- Rapid intravenous injection may cause vasoconstriction causing transient hypertension
- Consider cardiac monitoring in high risk patients requiring IV digoxin. See precautions for examples of high risk patients

9.2 Therapeutic drug monitoring:

- Not routinely required. Consider for patients at high risk of toxicity and on initiation of digoxin
- Following initiation or change in digoxin dose, steady state is reached after about 7 days (half-life is 36 hours)²
 - In patients with impaired renal function, the half-life of digoxin may be greatly prolonged. These patients take much longer to reach a steady state and their maintenance dose is lower. Digoxin plasma concentration should be monitored

- Take blood sample at least 6 hours after a dose (trough level) to allow for distribution
- The general therapeutic range is 0.5–2 microgram/L as toxicity is more common at digoxin concentrations >2 micrograms/L. However, toxic effects can occur at lower concentrations, particularly in the elderly or in those with electrolyte disturbance, hypoxia or hypothyroidism. Gastrointestinal symptoms (e.g. nausea, anorexia) may precede cardiac symptoms (e.g. arrhythmias).
- Consider maintaining lower concentrations between 0.5–0.8 microgram/L in patients with AF and in patients with heart failure who are in sinus rhythm as:
 - there may be improved outcomes or reduced hospitalisations with lower concentrations
 - higher concentrations within the therapeutic range may be associated with increased mortality risk
- Balance clinical benefit against adverse effects and dose reduce if appropriate.

10.0 Adverse effects^{1,2}

- Digoxin has a narrow therapeutic index. Adverse effects of digoxin are in general dose dependent and occur at doses higher than those needed to achieve a therapeutic effect.
- Refer to [SCGOPHCG approved references](#) (e.g. Australian Medicines Handbook) for more information.

Note: Digoxin may worsen arrhythmia due to proarrhythmic effects

TOXICITY

The clinical manifestations of chronic digoxin toxicity are non-specific. Digoxin intoxication commonly develops in the context of intercurrent illness, particularly where this leads to impaired renal function and digoxin elimination. It is often difficult to determine whether an unwell patient with an elevated serum digoxin level has digoxin toxicity or another cause for the observed clinical features

- Clinical features include nausea, vomiting and brady- or tachy-arrhythmias
- The significant risk is the development of malignant ventricular dysrhythmias (VT or VF)
- Cessation of digoxin therapy whilst evaluating plasma levels is warranted
- Digoxin specific antibody fragment (DigiFab®) is available for the treatment of life-threatening toxicity

The decision to administer digoxin specific antibody fragments involves an individualised risk assessment, and the Clinical Toxicology service at SCGH can give management advice if needed (contactable 24/7 via the hospital switchboard)

11.0 Compatibilities³

- Refer to the [Australian Injectable Drugs Handbook](#) for information.

12.0 Pregnancy and Breastfeeding⁴

- Safe to use in pregnancy and breast feeding
- Contact Obstetric Medicines Information Services phone: 08 6458 2723 for more information.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)
- NSQHS Standards: 1,4,5,6,8

References

1. MIMS Online [Internet]. Crows Nest (NSW): MIMS Australia Pty Ltd; 2023. Digoxin; [updated 2021 May 7; cited 2023 Aug 10]. Available from: MIMS Online
2. Australian Medicines Handbook [Internet]. Adelaide (SA): Australian Medicines Handbook Pty Ltd; Digoxin [cited 2023 Aug 10]. Available from: Australian Medicines Handbook.
3. Australian Injectable Drugs Handbook 9th ed [Internet]. Collingwood (VIC): The Society of Hospital Pharmacists of Australia. Digoxin [cited 2023 Aug 10]. Available from: Australian Injectable Drugs Handbook.
4. Therapeutics Guideline. [Internet]. Melbourne (VIC): Therapeutics Guidelines Ltd. Drug Use in Pregnancy and Breastfeeding; [cited 2023 Aug 10]

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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